QUBO Formulation: Database Transaction Serializability Verification

Database Transaction Serializability Verification

Problem Statement. Given a set of M candidate serial transaction orders, each associated with a pre-computed violation cost $c_j \geq 0$ that quantifies the number of dependency constraints violated by order j, the goal is to select exactly one candidate order such that the total violation cost is minimized. This corresponds to verifying whether the observed execution is serializable by identifying a candidate with zero violations, or otherwise determining the execution with the minimum number of violations.

Decision Variables. We introduce binary variables $x_j \in \{0, 1\}$ for each candidate order $j \in \{0, 1, \ldots, M-1\}$. The variable x_j indicates the selection of candidate order j, where $x_j = 1$ if order j is selected and $x_j = 0$ otherwise.

QUBO Derivation. The problem is formulated as a quadratic unconstrained binary optimization (QUBO) by encoding the objective and constraints into a single Hamiltonian $H(\mathbf{x})$.

Step 1: Objective Function. The objective is to minimize the total violation cost:

$$\min \sum_{j=0}^{M-1} c_j x_j. \tag{1}$$

This directly contributes to the QUBO Hamiltonian as:

$$H_{\text{obj}}(\mathbf{x}) = \sum_{j=0}^{M-1} c_j x_j. \tag{2}$$

Step 2: Constraint. Exactly one candidate order must be selected:

$$\sum_{j=0}^{M-1} x_j = 1. \tag{3}$$

Step 3: Penalty Term. We enforce the constraint using a quadratic penalty term with weight $\lambda > 0$:

$$H_{\text{pen}}(\mathbf{x}) = \lambda \left(\sum_{j=0}^{M-1} x_j - 1 \right)^2 . \tag{4}$$

Expanding the square and applying the idempotent property $x_j^2 = x_j$ for binary variables:

$$\begin{aligned} \left(\sum_{j=0}^{M-1} x_j - 1 \right)^2 &= \sum_{j=0}^{M-1} x_j^2 + \sum_{0 \le j \ne k \le M-1} x_j x_k - 2 \sum_{j=0}^{M-1} x_j + 1 \\ &= \sum_{j=0}^{M-1} x_j + \sum_{0 \le j \ne k \le M-1} x_j x_k - 2 \sum_{j=0}^{M-1} x_j + 1 \\ &= 1 - \sum_{j=0}^{M-1} x_j + \sum_{0 \le j \ne k \le M-1} x_j x_k. \end{aligned} \tag{5}$$

The double sum over distinct indices can be rewritten as twice the sum over ordered pairs $j < k$:

$$\sum_{0 \le j \ne k \le M-1} x_j x_k = 2 \sum_{0 \le j < k \le M-1} x_j x_k. \tag{6}$$

Substituting this back, the penalty term becomes:

$$\begin{aligned} H_{\text{pen}}(\mathbf{x}) &= \lambda \left(1 - \sum_{j=0}^{M-1} x_j + 2 \sum_{0 \le j < k \le M-1} x_j x_k \right) \\ &= \lambda - \lambda \sum_{j=0}^{M-1} x_j + 2\lambda \sum_{0 \le j < k \le M-1} x_j x_k. \end{aligned} \tag{7}$$

Step 4: Combined Hamiltonian. The total QUBO Hamiltonian is the sum of the objective and penalty terms:

$$\begin{aligned} H(\mathbf{x}) &= H_{\text{obj}}(\mathbf{x}) + H_{\text{pen}}(\mathbf{x}) \\ &= \sum_{j=0}^{M-1} c_j x_j + \lambda - \lambda \sum_{j=0}^{M-1} x_j + 2\lambda \sum_{0 \le j < k \le M-1} x_j x_k \\ &= \lambda + \sum_{j=0}^{M-1} (c_j - \lambda) x_j + 2\lambda \sum_{0 \le j < k \le M-1} x_j x_k. \end{aligned} \tag{8}$$

Step 5: Q Matrix and Offset. Reading off the coefficients from the general form $H(\mathbf{x}) = c + \sum_j Q_{j,j} x_j + \sum_{j<k} Q_{j,k} x_j x_k$, we obtain:

$$\begin{aligned} Q_{j,j} &= c_j - \lambda, \quad \forall j \in \{0, \ldots, M-1\}, \\ Q_{j,k} &= 2\lambda, \quad \forall 0 \le j < k \le M-1, \\ c &= \lambda. \end{aligned} \tag{9}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any infeasible solution has a strictly higher energy than any feasible solution. The minimum penalty incurred by any infeasible assignment is λ (occurring when $\sum x_j = 0$ or $\sum x_j \geq 2$). The maximum possible objective value for a feasible solution is $\max_j c_j$. Therefore, choosing

$$\lambda > \max_{j \in \{0,\ldots,M-1\}} c_j \tag{10}$$

guarantees that the energy of any infeasible solution exceeds the energy of any feasible solution, forcing the global minimum to be feasible.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0,1\}^M$ recovers the optimal solution because: (i) for all feasible solutions satisfying $\sum x_j = 1$, the penalty term vanishes and $H(\mathbf{x})$ equals the violation cost $\sum c_j x_j$; (ii) for all infeasible solutions, the penalty term is at least λ, and by the choice of λ, the energy of any infeasible solution is strictly greater than the maximum energy of any feasible solution. Consequently, the global minimum of $H(\mathbf{x})$ must be a feasible solution that minimizes the original objective.

Illustrative Example. Consider an instance with $M = 2$ candidate orders and violation costs $c_0 = 3$, $c_1 = 5$. We choose $\lambda = 10$, which satisfies $\lambda > \max\{3, 5\}$.

The QUBO parameters are:

$$\begin{aligned} Q_{0,0} &= 3 - 10 = -7, \\ Q_{1,1} &= 5 - 10 = -5, \\ Q_{0,1} &= 2 \times 10 = 20, \\ c &= 10. \end{aligned} \tag{11}$$

The Hamiltonian is:

$$H(x_0, x_1) = 10 - 7x_0 - 5x_1 + 20x_0x_1. \tag{12}$$

Evaluating H over all binary assignments:

$$\begin{aligned} &H(0,0) = 10, \quad \text{(infeasible, penalty 10)} \\ &H(1,0) = 10 - 7 = 3, \quad \text{(feasible, cost 3)} \\ &H(0,1) = 10 - 5 = 5, \quad \text{(feasible, cost 5)} \\ &H(1,1) = 10 - 7 - 5 + 20 = 18. \quad \text{(infeasible, penalty 10, cost 8)} \end{aligned} \tag{13}$$

The global minimum is $H(1,0) = 3$, corresponding to selecting candidate order 0 with minimum violation count 3.